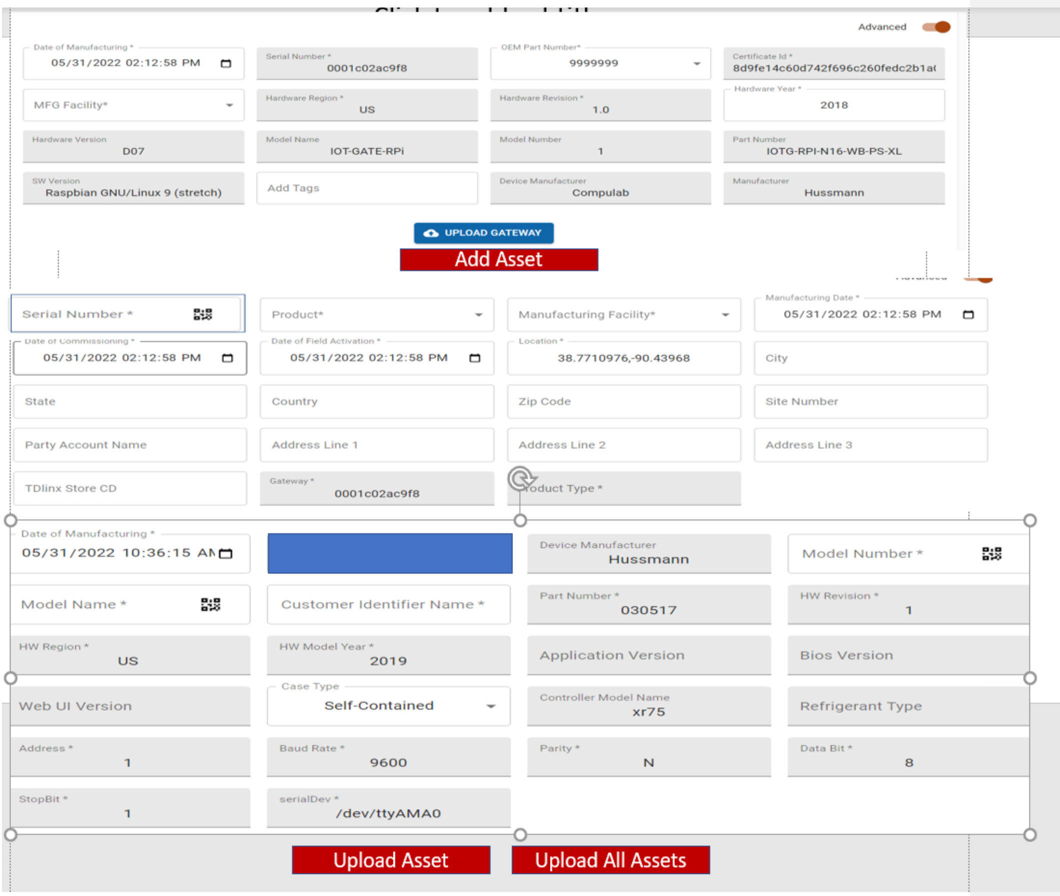
Config Tool Requirements For Production Phase

1. **Onboarding Page requirement**
    1. Need to remove Secondary Devices and add all the attributes from secondary to asset
    2. Onboarding page should look like as below



3. Remove Add Product and Upload ALL product buttons
4. Need to move all the parameters from Product to Asset as shown above
5. Case Type should be drop down list as below
6. Self-Contained
7. Remote Standalone
8. MDS
9. List of Controller – Controller type ,Model Name from Gateway
10. List of Product types from Gateway
11. From Controller Model Family and Product type need to populate product
12. Parameters like Location, City,State,Country ,Zip Code ,SiteNumber,Address Line1,Address Line2,Address Line3 and TDlinx Store ID should be editable for 1st Asset , but for the next assets need to copy the same as 1st and provide as editable feature
13. According to the Controller type communication parameters need to be populated

2. Config tool need to create automatic schema from product and controller reference file Below is the format for Schema

```
{
  "name": ,// short name of schema, not used by OneConnect
  "version": ,
  "desc": ,// description of schema
  "meta": {
    // add customer-specific attributes here, not used in OneConnect.
Naming is by customer.
    "connectionProtocol": "modbustcp",
    "controllerType": "" // e.g. "corelink","xr75"
  },

  "attributes":[
    {
      "id": "Night Curtain Override", // in telemetry message it will
be name of the attribute
      "desc": "Overrides night curtain position. Resets after 20 min.",
// can be used as info icon for remote configuration
      "write": true, // false/true
      "zone":"general" //"zone1", "zone2"
      "type": // types: boolean, string, integer, number
      "alias": [""], // attribute can have different name
      "unit": "pressure",
      "min": 80,
      "max": 725,
      "enum": [{   // for integer type enum section are applicable to
resolve int values to human-readable keys
          "val": 1,
          "key": "Upload"
        },{
          "val": 2,
          "key": "Download"
      }],
      "errorValue":[3660, -1]   // junk values sent by controller
      "factor": // DISCUSS

      "message":["locker_sensor_data", "locker_sensor_event"] // in
version 1 we support simple message list. In version 2 once
confiogurability willbe introduced, this section may be revisited
    }]
}
```

3. **Logs are not getting populated correctly on config tool**
    1. Need to resolve the logs issue
    2. Logs are not getting filtered correctly with the date
    3. Previous buttons are not working as expected

4. **Config tool to Support on all browsers**
    1. Config tool should be able to open on all the browsers
    2. For released version we are observing issue with IOS

5. **Config Tool need to support configuration for Gateway Wi-Fi**
    1. Config tool should have all the configurations required to set up Wi-Fi for gateway
    2. Right now below is the process we are following to configure Gateway but this needs to be automated using configtool

Wireless Enabling In Gateway

Install the access point and network management software

In order to work as an access point, the Raspberry Pi needs to have the hostapd access point software package installed:

```
sudo apt install hostapd
```

Enable the wireless access point service and set it to start when your Raspberry Pi boots:

```
sudo systemctl unmask hostapd
sudo systemctl enable hostapd
```

In order to provide network management services (DNS, DHCP) to wireless clients, the Raspberry Pi needs to have the dnsmasq software package installed:

```
sudo apt install dnsmasq
```

Finally, install netfilter-persistent and its plugin iptables-persistent. This utilty helps by saving firewall rules and restoring them when the Raspberry Pi boots:

```
sudo DEBIAN_FRONTEND=noninteractive apt install -y netfilter-persistent
iptables-persistent
```

Software installation is complete. We will configure the software packages later on.

Set up the network router

The Raspberry Pi will run and manage a standalone wireless network. It will also route between the wireless and Ethernet networks, providing internet access to wireless clients. If you prefer, you can choose to skip the routing by skipping the section "Enable routing and IP masquerading" below, and run the wireless network in complete isolation.

Define the wireless interface IP configuration

The Raspberry Pi runs a DHCP server for the wireless network; this requires static IP configuration for the wireless interface (wlan0) in the Raspberry Pi. The Raspberry Pi also acts as the router on the wireless network, and as is customary, we will give it the first IP address in the network: 192.168.4.1.

To configure the static IP address, edit the configuration file for dhcpcd with:

```
sudo nano /etc/dhcpcd.conf
```

Go to the end of the file and add the following:

```
interface wlan0
    static ip_address=192.168.4.1/24
    nohook wpa_supplicant
```

Commented [SVRRV1]: Add this in the dhcpcd.conf end of the file

Enable routing and IP masquerading

This section configures the Raspberry Pi to let wireless clients access computers on the main (Ethernet) network, and from there the internet. NOTE: If you wish to block wireless clients from accessing the Ethernet network and the internet, skip this section.

To enable routing, i.e. to allow traffic to flow from one network to the other in the Raspberry Pi, create a file using the following command, with the contents below:

```
sudo nano /etc/sysctl.d/routed-ap.conf
```

File contents:

```
# https://www.raspberrypi.org/documentation/configuration/wireless/access-point-routed.md
# Enable IPv4 routing
net.ipv4.ip_forward=1
```

Enabling routing will allow hosts from network 192.168.4.0/24 to reach the LAN and the main router towards the internet. In order to allow traffic between clients on this foreign wireless network and the internet without changing the configuration of the main router, the Raspberry Pi can substitute the IP address of wireless clients with its own IP address on the LAN using a "masquerade" firewall rule.

- The main router will see all outgoing traffic from wireless clients as coming from the Raspberry Pi, allowing communication with the internet.

- The Raspberry Pi will receive all incoming traffic, substitute the IP addresses back, and forward traffic to the original wireless client.

This process is configured by adding a single firewall rule in the Raspberry Pi:

```
sudo iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
```

Now save the current firewall rules for IPv4 (including the rule above) and IPv6 to be loaded at boot by the netfilter-persistent service:

```
sudo netfilter-persistent save
```

Filtering rules are saved to the directory /etc/iptables/. If in the future you change the configuration of your firewall, make sure to save the configuration before rebooting.

Configure the DHCP and DNS services for the wireless network

The DHCP and DNS services are provided by dnsmasq. The default configuration file serves as a template for all possible configuration options, whereas we only need a few. It is easier to start from an empty file.

Rename the default configuration file and edit a new one:

```
sudo mv /etc/dnsmasq.conf /etc/dnsmasq.conf.orig
sudo nano /etc/dnsmasq.conf
```

Add the following to the file and save it:

```
interface=wlan0 # Listening interface
dhcp-range=192.168.4.2,192.168.4.20,255.255.255.0,24h
                # Pool of IP addresses served via DHCP
domain=wlan     # Local wireless DNS domain
address=/gw.wlan/192.168.4.1
                # Alias for this router
```

The Raspberry Pi will deliver IP addresses between 192.168.4.2 and 192.168.4.20, with a lease time of 24 hours, to wireless DHCP clients. You should be able to reach the Raspberry Pi under the name gw.wlan from wireless clients.

There are many more options for `dnsmasq`; see the default configuration file (`/etc/dnsmasq.conf`) or the online documentation for details.

Commented [SVRRV2]: Add this information in dhsmasq.conf

Ensure wireless operation

Countries around the world regulate the use of telecommunication radio frequency bands to ensure interference-free operation. The Linux OS helps users comply with these rules by allowing applications to be configured with a two-letter "WiFi country code", e.g. US for a computer used in the United States.

In the Raspberry Pi OS, 5 GHz wireless networking is disabled until a WiFi country code has been configured by the user, usually as part of the initial installation process (see wireless configuration pages in this section for details.)

To ensure WiFi radio is not blocked on your Raspberry Pi, execute the following command:

```
sudo rfkill unblock wlan
```

This setting will be automatically restored at boot time. We will define an appropriate country code in the access point software configuration, next.

Configure the access point software

Create the hostapd configuration file, located at /etc/hostapd/hostapd.conf, to add the various parameters for your new wireless network.

```
sudo nano /etc/hostapd/hostapd.conf
```

Add the information below to the configuration file. This configuration assumes we are using channel 7, with a network name of NameOfNetwork, and a password AardvarkBadgerHedgehog. Note that the name and password should **not** have quotes around them. The passphrase should be between 8 and 64 characters in length.

```
country_code=US
interface=wlan0
ssid=HSMRasp01
hw_mode=g
channel=7
macaddr_acl=0
auth_algs=1
ignore_broadcast_ssid=0
wpa=2
wpa_passphrase=HSM1234567890
wpa_key_mgmt=WPA-PSK
wpa_pairwise=TKIP
rsn_pairwise=CCMP
```

Commented [SVRRV3]: Add this information in hostapd.conf, and here wifi

access point name is HSMRasp01 and Password HSM1234567890, you can change as per your requirements

Note the line country_code=GB:and IN for india it configures the computer to use the correct wireless frequencies in the United Kingdom. **Adapt this line** and specify the two-letter ISO code of your country. See Wikipedia for a list of two-letter ISO 3166-1 country codes.

To use the 5 GHz band, you can change the operations mode from hw_mode=g to hw_mode=a. Possible values for hw_mode are:

a = IEEE 802.11a (5 GHz) (Raspberry Pi 3B+ onwards)

b = IEEE 802.11b (2.4 GHz)

g = IEEE 802.11g (2.4 GHz)

Note that when changing the hw_mode, you may need to also change the channel - see Wikipedia for a list of allowed combinations.

Run your new wireless access point

Now restart your Raspberry Pi and verify that the wireless access point becomes automatically available.

sudo systemctl reboot

Once your Raspberry Pi has restarted, search for wireless networks with your wireless client. The network SSID you specified in file /etc/hostapd/hostapd.conf should now be present, and it should be accessible with the specified password.

If SSH is enabled on the Raspberry Pi, it should be possible to connect to it from your wireless client as follows, assuming the pi account is present: ssh pi@192.168.4.1 or ssh pi@gw.wlan

If your wireless client has access to your Raspberry Pi (and the internet, if you set up routing), congratulations on setting up your new access point

NOTE:

Once made all required steps, please provide the hostapd coniguration file full path to the **hostapd** for correct action.
For this reason please edit the */etc/default/hostapd* file by adding the following string:
DAEMON_CONF="/etc/hostapd/hostapd.conf